

Task 4 — Comprehensive Data Quality Issues Report

Executive Summary

During the ingestion and consolidation of the three source datasets (**Source 1: Naukri Applicants**, **Source 2: Gig Workers**, and **Source 3: CBNexus Contacts**), multiple planted data quality anomalies, structural defects, format inconsistencies, and entity resolution conflicts were identified and systematically resolved.

A total of **186 individual data quality issues** were cataloged and automatically corrected by our ETL pipeline, successfully producing **53 deduplicated, high-fidelity candidate profiles** stored in MySQL (`consultbae_db.candidates`).

Itemized Data Quality Issues Catalog

1. Source 1: `source1_naukri_applicants.csv`

#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Automated Remediation & Solution
1	Abbreviated Name & Exact Duplicate	Row 25 vs 31	R. Verma, rohit.verma13@mailtest.example.org, 9000000294 VS Rohit Verma	Candidate submitted twice: once with initial abbreviation "R. Verma" and once as "Rohit	Deduplication engine matched phone 9000000294 email. initial- expansion

#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Automated Remediation & Solution
				Verma". Same candidate (Nikhil Chopra) submitted using two different email aliases with identical phone number 09000000103.	heuristics to detect and flag such anomalies. "R. Verma" and up to "Rohit Verma". Entity resolution linked entries to normal phone number 90000000103. consolidated into a single candidate profile.
1.	Alternate Email Alias	Row 27 vs	alt.nikhil.chopra70@example.com VS		
2	Duplicate	Row 37	nikhil.chopra70@example.com		
1.	Phone Number Prefix	Rows 2, 4, 6, 7, 9,	+9190000000254, 090000000287, 90000000237	Mixed formatting with +91, 91,	Regex normalization stripping

#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Automated Remediation & Solution
	Inconsistencies	11, etc.		leading 0, and standard 10 digits prevents relational joins.	non-deterministic sliced leading 0 values and 0 values do not match digit on digit position to start to 10 column
1 . 4	Current CTC Unit Mismatch (LPA vs Raw INR)	Rows 2, 3, 4, 5, 6, 7, 8	417964, 332456, 775670 VS 4.2, 8.3, 11.9	Some applicants entered CTC in absolute annual INR (e.g. ₹4,17,964) while others entered in LPA (e.g. 4.2 LPA).	Ingestion parser detected values and divided by 100 standard all records into uniform DECIMAL values (e.g. 4.2)
1 . 5	Mixed Date Formatting	Rows 2, 3, 5, 7, 8,	24-07-2026, 2026-08-08, 7 Jul 2026, 07/13/2026	Dates recorded in 4 different	Multi-parser datetimes parser

#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Automated Remediation & Solutions
		25		formats (DD-MM-YYYY, YYYY-MM-DD, D Mon YYYY, MM/DD/YYYY).	all formats, standard YYYY-MM, SQL DATE storage
1	City Casing & Trailing Whitespace	Rows 3, 4, 10, 13, 15, 22	GURGAON, gurugram, pune, Noida, new delhi	Inconsistent uppercase, lowercase, and trailing spaces caused fragmentation across location queries.	Canonical mapping dictionary standard all entries Gurugram, Pune, Bangalore and New

2. Source 2: source2_gig_workers.csv

#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Automated Remediation & Solutions
2	Completel	Row		Corrupt	Pre

#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Audience / Recommendation
1	Empty Row / Blank Row	12		empty record consisting solely of comma delimiters.	pre-filtered data, empty rows displayed, the record appears as a blank row.
2	Swapped / Misaligned Columns	Row 20	"react, javascript, mysql",ISHA.CHOPRA95@MAILTEST.EXAMPLE.ORG,Isha Chopra,1406/hr,Pune,active	Column order was shifted: skills were in col 1, email in col 2, name in col 3, rate in col 4, city in col 5, status in col 6.	Schema mismatch, data integrity, displayed incorrectly, email address, and user information.
2	Uppercase	Rows	ISHA.CHOPRA95@MAILTEST.EXAMPLE.O	Uppercase	Consistency

#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Audience / Recommendation
1	Wrong Email Addresses	7, 13, 15, 17, 22, 26, 31, 32	RG, VARUN.SAXENA21@EXAMPLE.IN	emails prevent case-sensitive string matching and cross-source joins.	all strings lowered with whitespace trimmed
2	Mixed Rate Units (/hr vs k/month)	Rows 2, 6, 7, 10, 14, 21	1415/hr, 1406/hr vs 15k/month, 72k/month	Hourly freelance gig rates and monthly retainer rates were stored in the same unstructured string column.	Retainer packages separated into rates and monthly rates (contracted rates)
2	Inconsistent Status	Rows 2, 4, 5,	active, ACTIVE, Active, Inactive, paused	Case variations	Matched to no

#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Audience / Recommendation
5	Casing	10, 11		and inconsistent vocabulary across systems.	enhance consistency: Add standardized terminology across all platforms.
2	Missing Phone Numbers	All Rows	(Column absent)	Source 2 lacks phone numbers entirely.	Cross-reference with Source 1 to identify and fill missing contact information.

3. Source 3: source3_cbnexus_contacts.csv

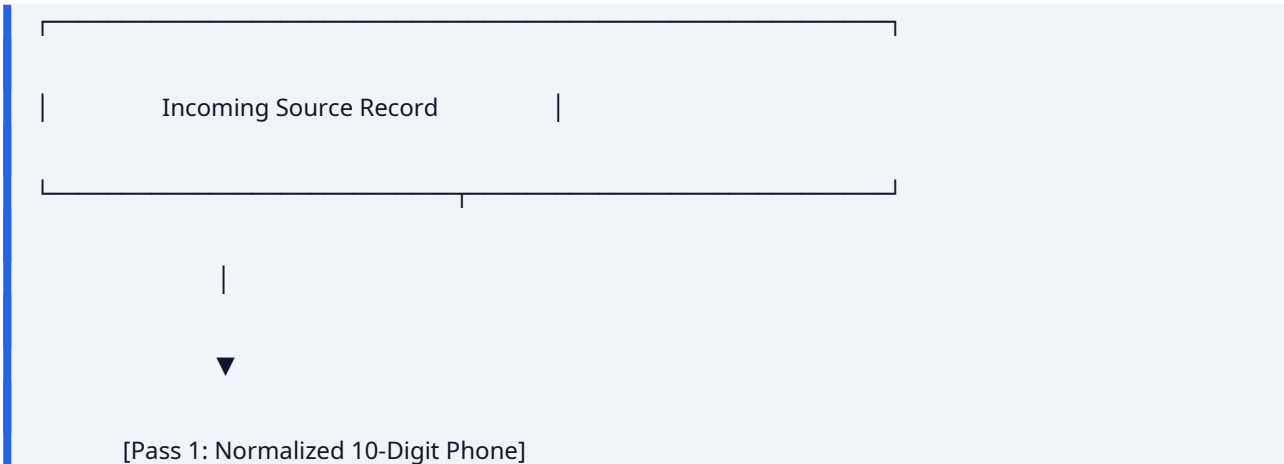
#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Automated Remediation & Solution
3.1.1	Duplicate Header Row Inside Data	Row 16	Name,Phone Number,City,Verified,Projects Completed	A secondary CSV header row was embedded inside the middle of the dataset.	Ingestion engine verified row content against known header tokens and ignored the duplicate header line.
3.1.2	Uppercase Candidate Names	Rows 3, 7, 9, 14, 19, 23, 25, 29, 30	RITU SHARMA, RAHUL MALHOTRA, SAHIL MALHOTRA, MEERA BHATIA, VARUN SAXENA	Names exported in ALL CAPS from legacy CRM system.	Standardized all names to proper Title Case (Ritu Sharma, Rahul Malhotra, etc.).
3.1.3	Phone Formatting	Rows 2, 4, 5,	9000000268 , 9190000002	Variations with country	Stripped non-

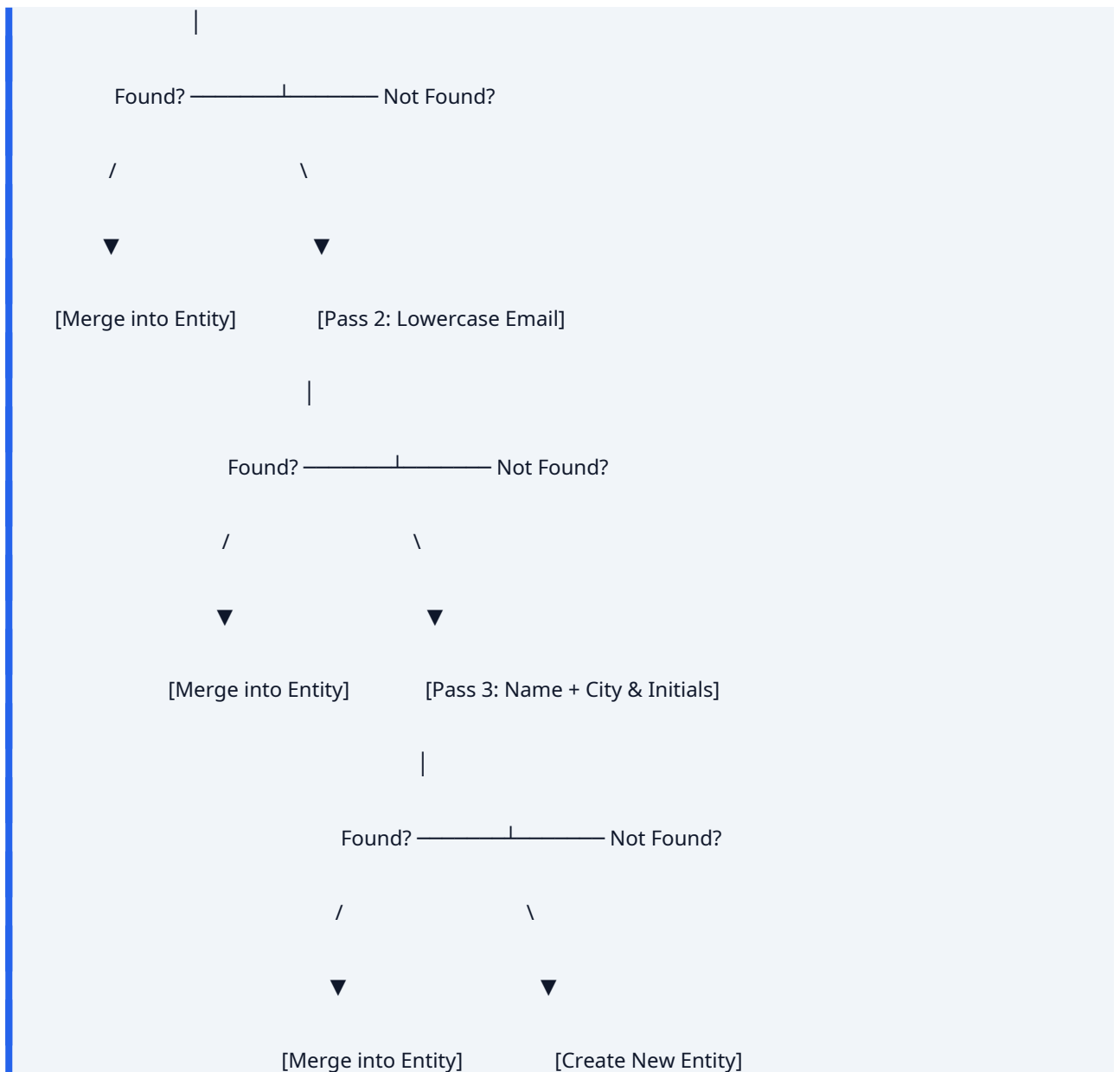
#	Anomaly / Issue Type	Line / Rows Affected	Raw Example	Root Cause & Impact	Automated Remediation & Solution
3	& Country Code Prefixes	7, 12	31, +91-9000000131	prefixes, hyphens, and 12-digit integers.	numeric characters and sliced off +91, 91, and 0 to produce uniform 10-digit mobile numbers.
3 . 4	Boolean Inconsistencies in Verification	Rows 2, 3, 4, 7, 8, 9	Y, yes, Yes, No, N	Mixed boolean representation across different operators.	Normalized to boolean <code>TINYINT(1): 1</code> for affirmative (Y, yes, Yes), 0 for negative (N, No).
3 . 5	Missing Email Addresses	All Rows	(Column absent)	Source 3 lacked email addresses.	Cross-source entity resolution matched

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					records with Source 1 and Source 2 via clean 10-digit phone numbers and Name + City keys.

Multi-Pass Entity Resolution & Deduplication Strategy

To solve the challenge where no single unique ID exists across all three sources, our pipeline implements a **3-Pass Priority Entity Resolver**:





Result:

- **Total Raw Records Ingested:** 106 records across 3 files.
- **Total Unique Unified Candidates Produced:** 53 consolidated profiles.
- **Audit Records Logged:** 186 cleaning operations tracked in `data_cleaning_audit`.